

NIKUNJ GARG

Java Backend Developer | BCA Student

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SUMMARY

Performance-driven **Java Backend Developer** specializing in scalable backend architectures, high-concurrency systems, and real-time distributed communication using **Java 21**, **Spring Boot 3**, and **PostgreSQL**. Experienced in designing low-latency **REST APIs**, multithreaded concurrency controls (pessimistic row locking, executor thread pools), **WebSockets (STOMP)**, and stateless **JWT** security pipelines across competitive hackathons and production services. Committed to clean architecture, ACID transactional guarantees, and automated testing (**JUnit 5**).

TECHNICAL SKILLS

- **Languages & Core Backend:** Java 21, SQL, Spring Boot 3, Spring Data JPA, Hibernate ORM, REST APIs, WebSockets (STOMP)
- **Database & Data Modeling:** PostgreSQL, Relational Schema Design (3NF), ACID Transactions, Indexing, Pessimistic Locking
- **Security & Architecture:** Spring Security, JWT (Refresh Token Rotation), BCrypt, Role-Based Access Control (RBAC), Clean Architecture
- **Testing & Concurrency:** JUnit 5, Concurrency Stress Testing (CountDownLatch, ExecutorService), Integration Testing, Mockito
- **DevOps, Cloud & Tools:** Docker, Docker Compose, Render Cloud, Git, GitHub, Maven, OpenAPI / Swagger 3.0, Postman, Cloudinary

PROJECTS

DevTinder | Java 21, Spring Boot 3, PostgreSQL, WebSockets (STOMP), Spring Security, Render [\[GitHub Repo\]](#) • [\[Live Demo\]](#)

- Prototyped the backend MVP during Devlynix Buildathon 2.0 with frontend peers; independently evolved and elevated the service into a production-grade system integrating with a Next.js client.
- Architected bidirectional real-time messaging using **WebSockets (STOMP)** paired with delta synchronization (`?after={id}`) and unread tracking backed by normalized **PostgreSQL** schemas, eliminating delivery latency.
- Engineered an **Algorithmic Skill Synergy Scoring engine** calculating 50–99% developer compatibility by weighting shared tech stacks, complementary skill disciplines, and hackathon project pitches.
- Hardened endpoints with stateless **JWT & Refresh Token Rotation**, sliding-window IP rate limiting (`RateLimitFilter`), incoming match radar (`/api/matches/requests`), and queue rewind (`/api/discover/reset-passes`).
- Containerized and deployed backend on **Render Cloud** with zero-downtime health monitors (`/api/health`), achieving 99.9% uptime and responsive sub-80ms API performance.

LibroSphere (High-Concurrency Asset Engine) | Java 21, Spring Boot 3, PostgreSQL, Spring Security, JUnit 5 [\[GitHub Repo\]](#)

- Eliminated race conditions and overselling on single-copy inventory under high contention by enforcing database row-level locking (`SELECT ... FOR UPDATE`) inside atomic transactions.
- Architected an automated FIFO reservation waitlist prioritizing queued members with a 48-hour pickup window before restoring uncollected inventory to general circulation.
- Engineered a scheduled cron reconciliation worker (`@Scheduled`) calculating tiered overdue penalties and added composite B-Tree indexes, slashing query latency from 120ms to under 15ms.
- Verified thread safety and ACID compliance with a **64-test automated JUnit 5 suite**, stress-testing 50 concurrent checkout threads via `CountDownLatch` with a 100% pass rate.

AgriSathi AI | Java 21, Spring Boot 3, PostgreSQL, Cloudinary CDN, Swagger 3.0 [\[GitHub Repo\]](#) • [\[Live Demo\]](#)

- Spearheaded the Spring Boot 3 backend for an agricultural intelligence platform during a 30-day hackathon, engineering 25+ RESTful endpoints serving crop health analytics to farmers and buyers.
- Orchestrated a multipart image pipeline streaming crop scans to an AI classification model with Cloudinary CDN storage, returning real-time disease diagnosis with confidence metrics.
- Enforced stateless **JWT authentication** with strict **Role-Based Access Control (RBAC)** to isolate farmer and buyer operations; documented and validated all endpoints via interactive **OpenAPI/Swagger 3.0**.

EDUCATION

Bachelor of Computer Applications (BCA) — Shri Guru Ram Rai University, Dehradun

2025 – 2028

Cumulative CGPA: 7.48 / 10.0 • **Semester 2:** 7.96 / 10.0

ACHIEVEMENTS & LEADERSHIP

- **Secretary — TechMantra Coding Club:** Promoted from Core Member to Secretary within 10 months; organized 5+ hands-on coding bootcamps and workshops for 120+ student developers.
- **Smart India Hackathon (SIH) Co-Lead:** Co-directed the university-wide internal evaluation rounds for SIH, mentoring and evaluating 20+ teams across software and hardware problem statements.
- **Startup Innovation Recognition:** Awarded Certificate of Recognition from **UCOST & SGRR University IIC** for architecting and pitching an innovative agritech startup solution.
- **Java Programming Certification:** Completed Java Programming certification through **IIT Bombay Spoken Tutorial**, validating OOP, multi-threading, and data structures.